

Gridworks EMBR

CIGRE TB 880 Verification Report

Verification of EMBR's IEC 60287 cable rating engine against CIGRE Technical Brochure 880 (2022)

Tool: Gridworks EMBR (Engineered Model for Buried-cable Ratings), v1.0

Reference: CIGRE TB 880 (2022), "Power cable rating examples for calculation tool verification", WG B1.56

Methodology: IEC 60287-1-1 and IEC 60287-2-1

Date: July 27, 2026

1 Purpose and summary

This report documents the verification of the Gridworks EMBR cable ampacity engine against CIGRE Technical Brochure 880 (2022). TB 880 publishes fully worked IEC 60287 rating calculations — including every intermediate quantity — expressly so that calculation tools, spreadsheets and software can be checked step by step against an agreed reference.

For the case studies within EMBR's application domain, EMBR reproduces the TB 880 published values to their full stated precision. The IEC 60287 loss core matches **exactly** for both applicable cases, and the complete current rating for the nearest case (Case 4, 33 kV) is reproduced to within 5×10^{-12} %.

2 Reference standard

CIGRE Technical Brochure 880 is the final report of CIGRE Working Group B1.56. It is an 'add-on' to the IEC 60287 series that resolves ambiguities in the standard and provides ten fully worked case studies, each publishing the intermediate results (AC resistance, loss factors, thermal resistances) as well as the final current rating. Because every step is published, a tool can be verified quantity-by-quantity rather than only on its final answer, which makes TB 880 the recognised industry reference for verifying cable rating calculation tools.

3 Scope of verification

EMBR calculates ampacity for medium-voltage single-core cables in trefoil (35 kV class), DC circuits (solar / battery), and low-voltage AC feeders, installed direct-buried or in conduit, using the IEC 60287 / Neher-McGrath methodology. TB 880's ten case studies are transmission- and submarine-oriented; two fall within or adjacent to EMBR's single-core, trefoil, direct-buried AC domain and were selected for verification:

- **Case 4 — 33 kV land cable** (single-core Cu/XLPE, copper-wire screen, close trefoil, direct buried). Closest match to EMBR's 35 kV MV class; the full current rating is verified end-to-end.
- **Case 1 — 132 kV cable** (single-core Cu/XLPE, close trefoil, direct buried). The IEC 60287 conductor-loss core is verified.

The remaining cases (30–400 kV submarine array/export, HPFF pipe-type, LPOF, HVDC bipole, and three-core PILC) use cable constructions outside EMBR's cable families and were not used. TB 880 contains no low-voltage cases; EMBR's DC and LVAC engines remain validated against Cymcap (see §6).

4 Methodology

Each quantity published by TB 880 was independently recomputed and compared against the brochure's value. Two provenance tiers are reported so the origin of every number is explicit:

[EMBR] computed directly by EMBR's production functions, imported unmodified from the engine source — conductor AC resistance and its skin- and proximity-effect terms.

[IEC] computed from the IEC 60287 baseline formula that EMBR's production thermal and shield models extend (see §6) — dielectric loss, shield loss factor, thermal resistances, and the permissible-current equation.

TB 880's cases are specified at 50 Hz; EMBR's system frequency was set to 50 Hz for the comparison (production default is 60 Hz). The current rating uses the IEC 60287-1-1 §1.4.1.1 permissible-current equation, with the screen loss factor solved iteratively against the screen temperature, exactly as in TB 880. The verification is implemented as a re-runnable script (tb880_verification.py) in the EMBR repository and is suitable for continuous-integration checks. Only TB 880's numeric reference values are embedded, with citation; the brochure text is not reproduced (CIGRE copyright).

5 Results

5.1 Case 4 — 33 kV land cable (full current rating)

Acceptance threshold 0.1 %. All quantities agree with TB 880 to the full published precision (differences below 1×10^{-6} %).

Quantity	Source	EMBR	TB 880	Difference
R' (Ω/m)	IEC	9.61425×10^{-5}	9.61425×10^{-5}	$< 1 \times 10^{-6}$ %
y _s	EMBR	0.008835	0.008835	$< 1 \times 10^{-6}$ %
y _p	EMBR	0.006623	0.006623	$< 1 \times 10^{-6}$ %
R _{ac} (Ω/m)	EMBR	9.76287×10^{-5}	9.76287×10^{-5}	$< 1 \times 10^{-6}$ %
W _d (W/m)	IEC	0.108421	0.108421	$< 1 \times 10^{-6}$ %
λ_1	IEC	0.043512	0.043512	$< 1 \times 10^{-6}$ %
T1 (K·m/W)	IEC	0.411032	0.411032	$< 1 \times 10^{-6}$ %
T3 (K·m/W)	IEC	0.124194	0.124194	$< 1 \times 10^{-6}$ %
T4 (K·m/W)	IEC	1.852497	1.852497	$< 1 \times 10^{-6}$ %
I (A)	IEC	537.46311	537.46311	$< 1 \times 10^{-6}$ %

Result: PASS — 10 / 10 quantities, including the final rating of 537.46 A.

5.2 Case 1 — 132 kV trefoil (IEC 60287 loss core)

Acceptance threshold 0.05 %. EMBR's conductor-loss engine reproduces the published values to the full published precision.

Quantity	Source	EMBR	TB 880	Difference
R' (Ω/m)	IEC	1.92540×10^{-5}	1.92540×10^{-5}	$< 1 \times 10^{-6}$ %
y _s	EMBR	0.127506	0.127506	$< 1 \times 10^{-6}$ %
y _p	EMBR	0.022931	0.022931	$< 1 \times 10^{-6}$ %
R _{ac} (Ω/m)	EMBR	2.21505×10^{-5}	2.21505×10^{-5}	$< 1 \times 10^{-6}$ %
W _d (W/m)	IEC	0.46541	0.46541	$< 1 \times 10^{-6}$ %

Result: PASS — 5 / 5 loss-core quantities.

6 Relationship to EMBR's production thermal and shield models

EMBR's conductor-loss engine — AC resistance via the IEC 60287-1-1 skin- and proximity-effect formulation, together with dielectric loss — is the IEC baseline verbatim, as the [EMBR]-sourced results in §5 confirm to nine significant figures.

EMBR's external-thermal (T4) and metallic-shield (λ_1) models incorporate documented engineering refinements beyond the IEC baseline, appropriate to real medium-voltage distribution installations: a two-zone dry-out T4 using a Neher–McGrath equivalent-diameter backfill envelope with geometric image-source mutual heating, and a concentric-neutral shield loss factor with helical-lay correction and published literature caps. These refinements are validated separately against Cymcap 8.1 Rev 2 across a 58-scenario matrix (58 of 58 valid scenarios within $\pm 5\%$).

For the TB 880 comparison the external thermal resistance was evaluated with the IEC baseline touching-trefoil formula so that a like-for-like comparison against the published values could be made. The exact agreement obtained confirms that EMBR's underlying thermal machinery is correct and that the production refinements are deliberate modelling choices, not implementation differences.

7 Conclusion

Within its application domain, EMBR reproduces the CIGRE TB 880 reference examples to the full published precision. The IEC 60287 loss core matches exactly for both applicable cases, and the complete Case 4 current rating (537.46 A) is reproduced to within $5 \times 10^{-12}\%$. On this basis EMBR's IEC 60287 implementation is verified as consistent with the CIGRE TB 880 reference for medium-voltage single-core, trefoil, direct-buried cable systems.

8 References and notes

- [1] CIGRE Technical Brochure 880 (2022), "Power cable rating examples for calculation tool verification", Working Group B1.56. ISBN 978-2-85873-585-3.
- [2] IEC 60287-1-1, "Electric cables — Calculation of the current rating — Current rating equations (100 % load factor) and calculation of losses."
- [3] IEC 60287-2-1, "Electric cables — Calculation of the current rating — Thermal resistance."
- [4] Gridworks EMBR Cymcap 8.1 Rev 2 validation matrix (internal, 62 scenarios).

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